FORENSIC ANALYSIS REPORT

Blood Pattern Analysis & Classification

Report Generated: March 10, 2026 at 12:42 PM

EXECUTIVE SUMMARY

Analysis Component	Result	Confidence
Blood Detection	■ LIKELY BLOOD	90.2%
Weapon Type	Gun	99.8%
Point of Origin	(320.7, 235.0)	337 droplets

1. BLOOD DETECTION ANALYSIS

Verdict: ■ LIKELY BLOOD

Confidence: 90.2%

Color Analysis:

Coverage: 27.40%

Score: 91.3%

Detected Blood Types:

• Fresh Blood (Bright Red): 16.58%

• Fresh Blood (Deep Red): 9.89%

• Dried Blood (Brown-Red): 5.94%

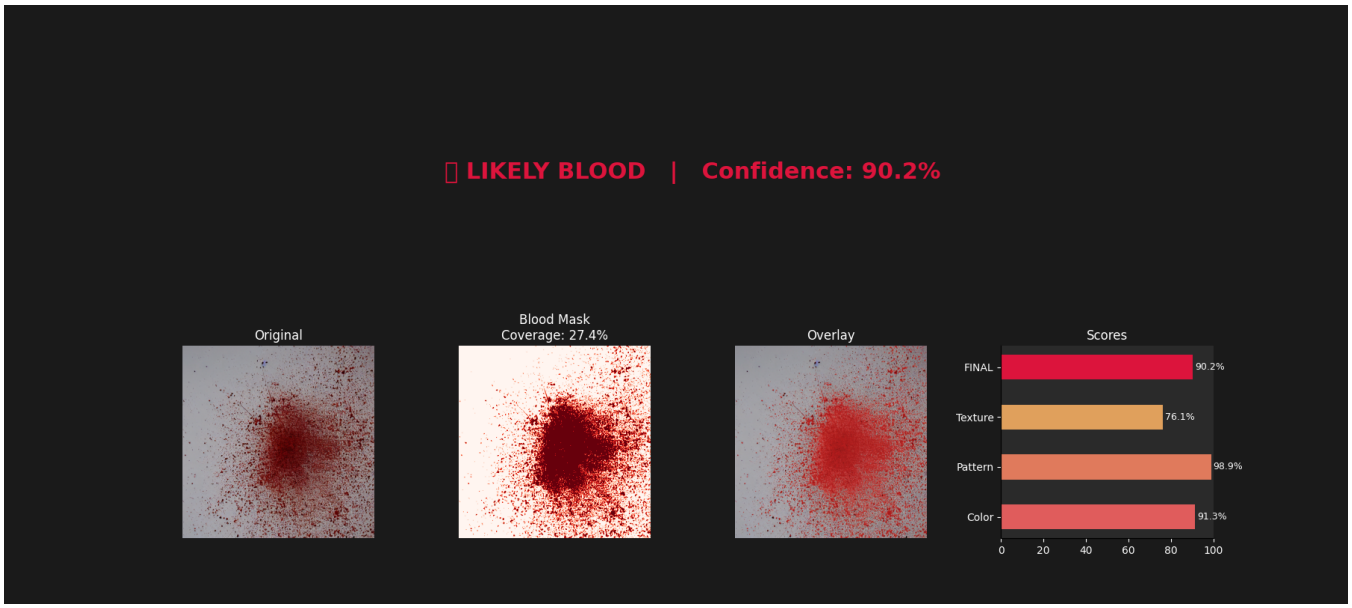
Pattern Analysis:

Score: 98.9%

Components: 549

Satellites: 548

Blood Detection Visualization



Blood Detection Analysis

2. WEAPON TYPE CLASSIFICATION

Weapon Type: Gun

Confidence: 99.8%

Detailed Probabilities:

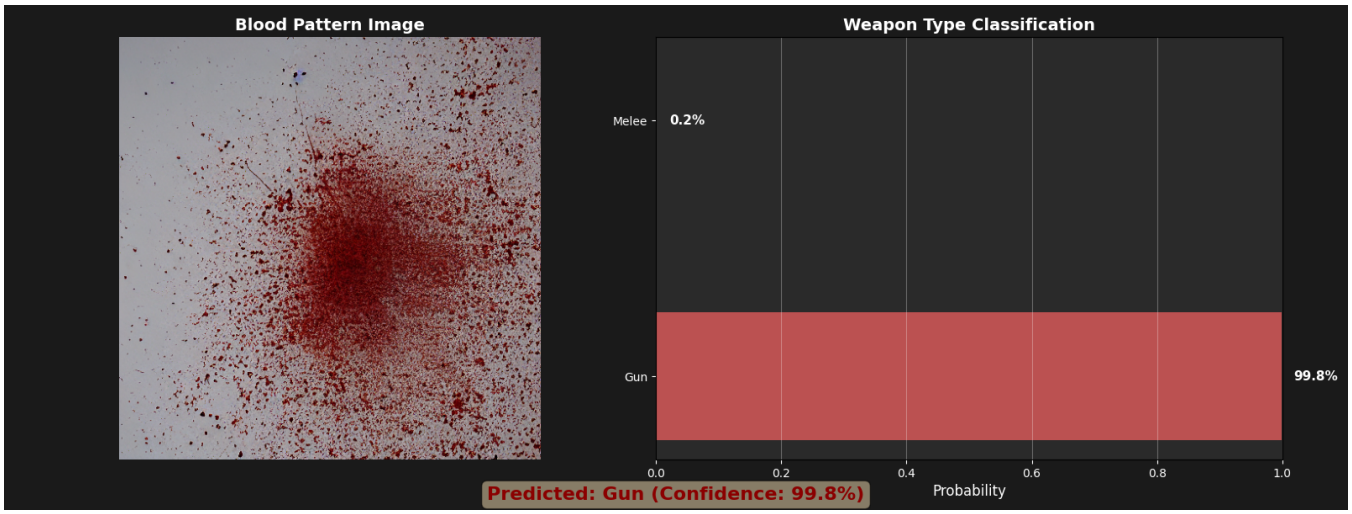
• Gun: 99.85%

• Melee: 0.15%

Interpretation:

High/medium velocity impact spatter detected. Characteristics: Fine mist, small circular droplets, uniform distribution typical of gunshot wounds.

Weapon Classification Visualization



Weapon Classification Analysis

3. POINT OF ORIGIN ANALYSIS

Point of Origin: (320.7, 235.0)

Analysis Statistics:

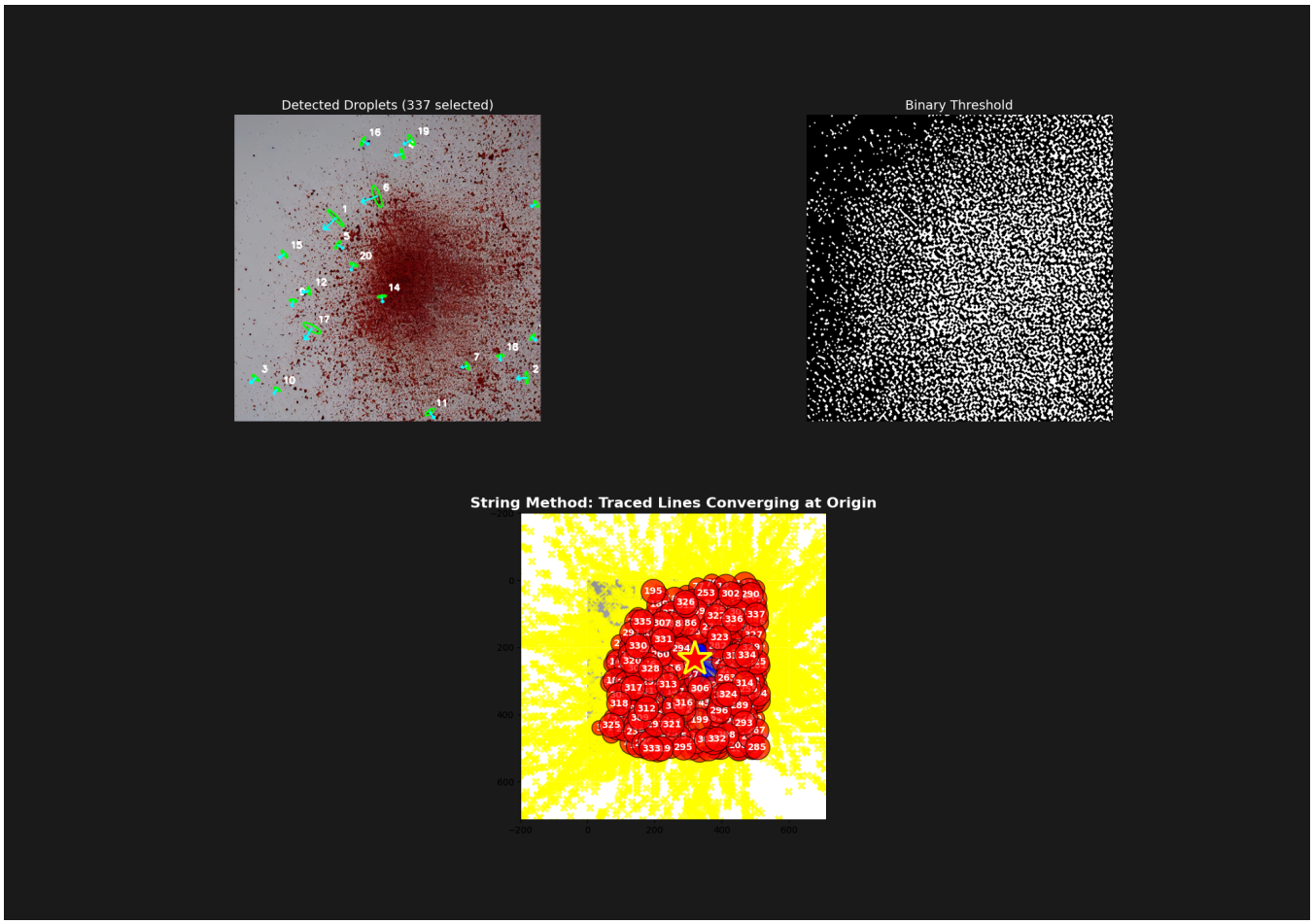
Droplets Analyzed: 337

Intersection Points: 29235

Average Impact Angle: 35.0°

Angle Range: 9.7° - 56.4°

Point of Origin Visualization



String Method Analysis - Trajectory lines and origin point

DISCLAIMER

This forensic analysis report was generated using AI-assisted computational methods for blood pattern analysis. The results presented are based on image analysis algorithms and machine learning models trained on forensic datasets. **Important Notes:**

• This report should be used as a supplementary tool in forensic investigations
• Results must be verified by qualified forensic experts
• This report is not intended for sole use in legal proceedings
• Professional forensic confirmation is required for official documentation
• The accuracy of results depends on image quality and pattern clarity **Model Information:**
• Blood Detection: Computer vision analysis (HSV color space, pattern recognition)
• Weapon Classification: Deep learning CNN (TensorFlow)
• String Method: Classical forensic trajectory analysis with DBSCAN clustering For questions or concerns about this analysis, consult with certified forensic analysts.